

Supplementary Information

An open-source, 3D-printed auger powder doser designed with generative artificial intelligence

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S1 Bill of materials

Table S1 lists the complete per-channel bill of materials for the single-channel module, plus system-shared items (one per platform regardless of channel count). Prices are single-unit USD as of 2026-Q2 and should be reconfirmed at order time; vendor links and mirrored vendor design files are maintained in `hardware/` in the repository.

Table S1: Bill of materials. Printed parts are excluded (about 350 g PLA per channel, <\$10), as is the gravimetric-feedback instrument (A&D HR-100A analytical balance, 0.1 mg readability), shared laboratory equipment that most labs already own.

Part	Qty	Price (USD)	Source
<i>Per channel</i>			
NEMA-11 bipolar stepper, 0.67 A, 10 N·cm (11HS18-0674S)	1	14	StepperOnline
Pololu Tic T500 stepper controller	1	33	Pololu #3135
JF-0530B 5 V push-pull solenoid (tap actuator)	1	7.50	Adafruit #412
DRV8871 H-bridge breakout (solenoid drive)	1	7.50	Adafruit #3190
ERM coin vibration motor, 10 mm	1	1.95	Adafruit #1201
DRV2605L haptic driver breakout (I ² C)	1	7.95	Adafruit #2305
MG996R-class metal-gear servo (tilt)	1	12	generic
6805ZZ bearing (auger support)	1	3	generic
5 mm shaft coupler, fasteners (M3/M5), bulk capacitors	—	10	generic
<i>System-shared (one per platform)</i>			
Raspberry Pi Pico W	1	6	Raspberry Pi
MAX3232 RS-232 transceiver breakout (balance interface)	1	3	generic
Mean Well GST60A12-P1J 12 V/5 A supply + cord + pigtail	1	26	Digi-Key
Pololu D24V22F5 5 V buck regulator	1	19	Pololu #2858
Pololu 33 V shunt regulator (back-EMF clamp)	1	15	Pololu #3776
Breadboard/protoboard, headers, wiring	—	10	generic
Total, single-channel platform		≈175	

S2 Construction guide

A step-by-step illustrated build guide is maintained in the repository (<https://github.com/vertical-cloud-lab/powder-doser>) alongside the parametric CAD sources and STLs for every part. In outline:

1. **Print** the auger, gear pair, bracket, tap collar (two clamp halves), mounting plate, hinged baseplate, and servo sector in PLA (0.2 mm layers, 4 perimeters, 25% infill, no supports; print the auger vertically, helix up).

2. **Auger drive:** press the 6805ZZ bearing into the bracket, seat the auger, mesh the printed gear pair, and bolt the NEMA-11 stepper to the front face of the mounting plate.
3. **Tap collar:** clamp the two collar halves around the auger tube; mount the solenoid to the collar flange (2×M3) so the plunger strikes the anvil pad, and adhere the ERM motor to the opposite pad.
4. **Tilt stage:** join the mounting plate to the baseplate through the side hinges (printed pins), engage the servo sector, and confirm the exit nozzle lies on the hinge axis.
5. **Electronics:** wire the Pico W, Tic T500 (UART), DRV8871, DRV2605L (I²C), MAX3232 (RS-232 to the HR-100A balance), and servo per the KiCad schematic in `hardware/test-module/`; flash the MicroPython firmware.
6. **Checkout:** run the firmware self-test (rotate, tap, vibrate, tilt, tare/read), then a per-powder calibration sweep.

S3 Powder test protocols

Per-powder characterization runs a uniform test battery: one fixed sequence of seven *test protocols*, executed identically for every powder so that the results are directly comparable across powders (Table S2). The battery is deliberately a characterization sequence rather than an optimization workflow—every parameter is frozen, and the object is to observe how each powder responds under identical conditions.

Protocols A–F isolate the platform’s degrees of freedom one at a time. Protocol A establishes the balance noise floor for the run, so that every subsequent mass change is read against a measured floor rather than an assumed one. Protocol B holds each tilt angle with no actuator commanded, which separates powders that discharge under gravity alone from those that require actuation, and which measures the horizontal-park shutoff directly. Protocols C and D characterize the auger itself, respectively as mass delivered per revolution at each tilt and as mass flow rate versus rotation speed. Protocols E and F characterize the two agitation actuators, each trial pairing a metered re-feed rotation with a single actuation burst so that the recorded mass change is attributable to the burst alone. Protocol G then exercises the three-phase closed-loop controller described in the Experimental section of the main text, which is built on those primitives.

Each protocol emits machine-readable trial records over the serial link, captured as versioned CSV and stamped with the powder identifier, the battery version, and the run parameters, so that every reported number is traceable to its raw balance stream. The firmware and the archived CSVs identify the protocols by the same letters used here. The battery implementation, its parameter defaults, and a hardware-independent test suite are in `hardware/test-module/firmware/` in the repository.

S4 Exit-nozzle variants

Four printed exit-nozzle geometries were produced for dispense testing (Fig. S1); the dispensing campaign will report flow-initiation and dribble behaviour per nozzle and per powder.

S5 AI usage log

In line with Digital Discovery’s guidance on LLM use, all AI interactions are preserved verbatim and publicly:

- **CAD/code generation:** every agent session (GitHub Copilot coding agent; Claude-family models, April–June 2026) is recorded as a pull request containing the prompt (issue text and review comments), the generated CAD code, rendered views, and the human review thread.
- **Design reviews and literature queries:** Edison Scientific query scripts, task identifiers, raw answers, and extracted references are archived under `paper/background/` on the corresponding branches; the consolidated, validated bibliography with per-entry verification status is at `paper/references_validation.md`.
- **Text-to-CAD evaluations:** CADSmith run artifacts and the Zoo Design Studio (Zookeeper agent) session transcripts and best-practice notes are preserved on their respective branches.
- **Qualitative observations:** the running team journal of AI behaviour (good and bad) is public in the repository’s discussion #39.

Table S2: Uniform powder test battery. Seven test protocols are run in a fixed order with frozen parameters on every powder, so that the results are directly comparable across powders; the battery is a characterization sequence rather than an optimization workflow. Protocols A–F characterize the actuation primitives one at a time, and protocol G exercises the three-phase closed-loop controller built on them. Tilt is the auger tube’s angle above horizontal, from 0° (horizontal park) to 45°, the steepest tilt tested. *Trials* is the number of machine-readable trial records each protocol emits per run, which is the unit of replication in the analysis. Parameter values are the frozen defaults of the battery firmware (`hardware/test-module/firmware/powder_battery.py`), which identifies the protocols by the same letters. The firmware and its raw records give tilt as a servo setting (0, 45, or 90), which the 2:1 tilt gear turns into tube angles of 0°, 22.5°, and 45°.

Protocol	Varied factor	Fixed parameters	Quantity measured	Trials
A Balance baseline	Nothing	Tilt 22.5°; 8 reads	Mass change with no actuator commanded, giving the per-run noise floor every other protocol is read against	8
B Static hold	Tilt (0/22.5/45°)	15 s hold; no actuation	Spontaneous discharge under gravity alone, separating free-flowing powders from those needing actuation and testing the horizontal-park shutoff claim	3
C Rotation yield	Tilt (0/22.5/45°)	6 × 360° steps; 30 rpm	Mass delivered per auger revolution and its revolution-to-revolution spread, i.e. the feed factor and precision at each tilt	18
D Speed sweep	Auger speed (15/45/90 rpm)	Tilt 22.5°; 3 rev continuous; balance streamed every 250 ms	Mass flow rate versus rotation speed, and the within-revolution pulsation of the discharge from the streamed mass trace	3 (+ polls)
E Tap yield	Tilt (0/22.5°)	8 trials × (360° re-feed + 1 solenoid tap, 60 ms on)	Mass released per solenoid tap, with the metered re-feed rotation logged separately so the tap delta is tap-only	32
F Vibration yield	Tilt (0/22.5°)	As protocol E, with 3 ERM bursts replacing the tap	Mass released per vibration burst, the counterpart of protocol E for the eccentric-rotating-mass actuator	32
G Closed-loop dose	Nothing	3 doses; 1.000 g target; three-phase controller	Delivered mass, dose error, time to dose, and the bulk/fine/tap cycle breakdown of the closed-loop controller	3 doses

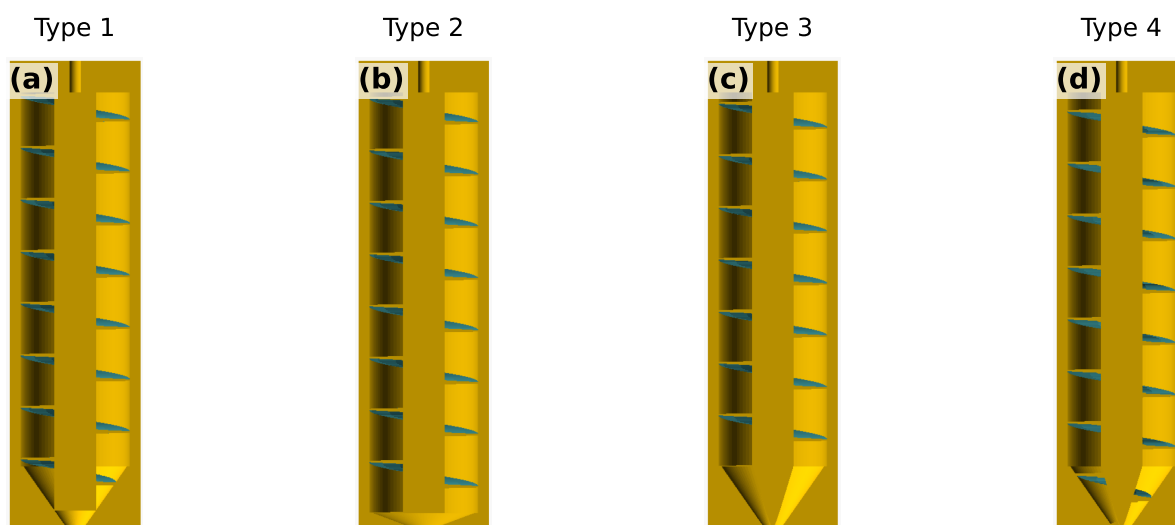


Figure S1: Cross-sections of the four exit-nozzle variant CAD models (Types 1–4, panels (a)–(d)) of the test auger, differing in exit constriction and chamfer geometry. The variants are parametric OpenSCAD models (`cad/auger-geared/nozzle-variants.scad`) authored by AI coding agents under human review; no GUI CAD package was used.